

# Pearson Correlation and Interpretation

## Important Concept: Pearson Correlation Coefficient

The **Pearson correlation coefficient (r)** is a statistical measure used to determine the **strength and direction of the linear relationship between two numerical variables**.

The value of **r** always lies between **-1 and +1**.

| Value of r   | Interpretation               |
|--------------|------------------------------|
| +1           | Perfect positive correlation |
| $0 < r < +1$ | Positive correlation         |
| 0            | No linear correlation        |
| $-1 < r < 0$ | Negative correlation         |
| -1           | Perfect negative correlation |

## Pearson Correlation Formula

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Where:

- $x_i$  = individual value of variable X
- $y_i$  = individual value of variable Y
- $\bar{x}$  = mean of X
- $\bar{y}$  = mean of Y
- $r$  = Pearson correlation coefficient

## Interpretation

- **Positive correlation:** As X increases, Y generally increases.
- **Negative correlation:** As X increases, Y generally decreases.
- **Zero correlation:** There is no clear linear relationship between X and Y.

**Important:** Correlation shows association, not necessarily causation.

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# Question 1

## Instagram Usage and Mobile Data Usage

Suppose the following data is collected from 8 friends.

- **X = Hours spent on Instagram per week**
- **Y = Monthly mobile data usage in GB**

### Given Data

| Friend | Instagram Hours/Week (X) | Mobile Data Usage (Y) in GB |
|--------|--------------------------|-----------------------------|
| 1      | 5                        | 3                           |
| 2      | 7                        | 5                           |
| 3      | 8                        | 6                           |
| 4      | 10                       | 8                           |
| 5      | 12                       | 9                           |
| 6      | 14                       | 11                          |
| 7      | 15                       | 12                          |
| 8      | 18                       | 14                          |

### Step 1: Calculate the Means

$$\bar{x} = \frac{5+7+8+10+12+14+15+18}{8} = \frac{89}{8} = 11.125$$

Therefore, the mean Instagram usage is:

**11.125 hours/week**

For mobile data:

$$\bar{y} = \frac{3+5+6+8+9+11+12+14}{8} = \frac{68}{8} = 8.5$$

Therefore, the mean mobile data usage is:

**8.5 GB/month**

### Step 2: Calculate Pearson Correlation

Using the Pearson correlation formula:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Using a calculator or spreadsheet:

$$r \approx 0.9973$$

## Interpretation

The correlation coefficient is approximately **+0.997**.

Since the value is very close to **+1**, there is a **very strong positive correlation** between weekly Instagram usage and monthly mobile data usage in this sample.

This means that friends who spend more time on Instagram generally tend to use more mobile data.

## Conclusion

**There is a very strong positive linear correlation between Instagram usage and mobile data usage.**

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# Question 2

## Python Function to Calculate Pearson Correlation

The following function calculates the Pearson correlation coefficient using only the basic `math` module from the Python standard library.

```
import math
```

```
def compute_correlation(x, y):
```

```
    if len(x) != len(y):
```

```
        return "Lists must have equal length"
```

```
    mean_x = sum(x) / len(x)
```

```
    mean_y = sum(y) / len(y)
```

```
    numerator = sum((x[i] - mean_x) * (y[i] - mean_y) for i in range(len(x)))
```

```
    sum_x = sum((x[i] - mean_x) ** 2 for i in range(len(x)))
```

```
    sum_y = sum((y[i] - mean_y) ** 2 for i in range(len(y)))
```

```
    denominator = math.sqrt(sum_x * sum_y)
```

```
if denominator == 0:  
    return "Correlation cannot be calculated"
```

```
return numerator / denominator
```

### Example

```
x = [5, 7, 8, 10, 12, 14, 15, 18]
```

```
y = [3, 5, 6, 8, 9, 11, 12, 14]
```

```
print(compute_correlation(x, y))
```

### Output

```
0.9972565106055741
```

Approximately:

```
r=0.9973\boxed{r=0.9973}
```

### Explanation of the Function

1. The function first checks whether both lists have the same number of values.
2. It calculates the mean of X and Y.
3. It calculates the numerator of the Pearson formula.
4. It calculates the squared deviations for X and Y.
5. `math.sqrt()` calculates the square root required in the denominator.
6. Finally, it returns the Pearson correlation coefficient.

### Interpretation

Since:

```
r≈+0.997r \approx +0.997
```

the two variables have a **very strong positive correlation**.

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## Question 3

### IMDb Ratings and BookMyShow Reviews

## Objective

The purpose is to examine whether there is a relationship between:

- **IMDb rating of a movie**
- **Number of reviews on BookMyShow**

For an assignment demonstration, the following 10-movie dataset can be entered into **Excel**, **Google Sheets**, or **Python**. Movie-platform figures can change over time, so the values should be treated as a sample dataset rather than permanent values.

| No. | Movie   | IMDb Rating | BookMyShow Reviews |
|-----|---------|-------------|--------------------|
| 1   | Movie A | 8.5         | 8,500              |
| 2   | Movie B | 8.1         | 7,200              |
| 3   | Movie C | 7.9         | 6,800              |
| 4   | Movie D | 7.6         | 5,900              |
| 5   | Movie E | 7.4         | 5,100              |
| 6   | Movie F | 7.1         | 4,300              |
| 7   | Movie G | 6.9         | 3,900              |
| 8   | Movie H | 6.6         | 3,100              |
| 9   | Movie I | 6.3         | 2,500              |
| 10  | Movie J | 6.0         | 2,000              |

## Calculation

The Pearson correlation coefficient can be calculated in Excel or Google Sheets using:

`=CORREL(C2:C11,D2:D11)`

Using the above sample data, the correlation is approximately:

$r \approx 0.99$

## Interpretation

The correlation coefficient is **positive and very strong**.

This means that, in this sample dataset, movies with higher IMDb ratings tend to have a higher number of BookMyShow reviews.

However, this does **not** mean that a higher IMDb rating causes more BookMyShow reviews. Other factors such as movie popularity, marketing, number of viewers, star cast, release scale, and audience interest may also affect the number of reviews.

## Conclusion

**The sample shows a strong positive correlation between IMDb rating and the number of BookMyShow reviews.**

**Data note:** BookMyShow states that users can rate or review a movie after booking tickets through BookMyShow, so the review count is related to user activity on the platform.

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## Question 4

### Zomato Food Orders and IPL Matches

Suppose there is a strong positive correlation between the number of food orders on Zomato and the number of IPL matches aired in a month.

A strong positive correlation **does not prove that IPL matches cause more food orders**. Other factors such as weekends, holidays, promotions, weather, major matches, discounts, and increased overall customer activity could influence both variables.

Therefore, **correlation does not necessarily mean causation**, and additional analysis or a controlled experiment would be required to establish a causal relationship.

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## Key Points to Remember

### Pearson Correlation

$$-1 \leq r \leq +1$$

- $r$  close to  $+1$  → Strong positive correlation
- $r$  close to  $-1$  → Strong negative correlation
- $r$  close to  $0$  → Little or no linear correlation

### Important Difference

**Correlation  $\neq$  Causation**

A correlation tells us that two variables are related, but it does not by itself prove that one variable causes the other.

## **Practical Example**

**Instagram usage**  $\uparrow \rightarrow$  **Mobile data usage**  $\uparrow$

This can show positive correlation.

But:

**IPL matches**  $\uparrow \rightarrow$  **Zomato orders**  $\uparrow$

does not automatically prove that IPL matches caused the increase in orders.

Other variables may be responsible for the observed relationship.